ADOBE – STRONGER THAN CONCRETE?

Field Experiments in Biological, Self-Cleaning, and Energy-Efficient Adobe Construction

Author: István Simor | **ORCID:** 0009-0002-6599-3480 | **Date:** 2026-03-17 | **Version:** v1.0

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ABSTRACT

This technical paper documents nearly two decades of experimental restoration and field observation of an adobe house on the author's property in Hungary, under controlled conditions. The purpose of this work is not to provide architectural instructions but to record material behavior and innovative applications.

The restoration employs a **biological frame structure** of acacia stakes and woven branch lattices into which a **fermented binder** is injected. The binder consists of recycled adobe, clay, sand, whey, and fresh horse manure. Anaerobic bacteria produce cellulose solutions that act as a natural biopolymer adhesive after drying, without cement or synthetic resins.

Key measurement: During external temperatures as low as −22 °C, water inside the unheated building did not freeze, demonstrating the combined thermal mass of gravel floors and adobe walls, together with geothermic buffering.

The study also documents the development of **whey–manure stabilized adobe panels** (4 cm thick, exceptionally strong, heat-storing, biodegradable). This publication contributes empirical case studies to the EFU (Ecological Flux Unit) system L1 Praxis layer for circular, zero-waste construction.

Keywords: adobe, fermented binder, whey, horse manure, cellulose biopolymer, thermal mass, circular material use, adobe panel innovation, EFU.

I. INTRODUCTION: THE SPARKLING EARTH

Nearly five decades in the construction sector have shaped my understanding of materials. Initially, I believed in the dominance of concrete and modern technologies. The adobe walls of my house, however, revealed their true potential: under the chisel, the material **sparked**.

Adobe, when understood and correctly managed, is one of the most noble and durable building materials. This study aims to preserve that knowledge. It is **not** a building manual—construction should be undertaken only by trained professionals.

II. SYSTEMS APPROACH: HALTING ENTROPY

Adobe house degradation is **not inevitable** but results from poor moisture and vapor management. Successful restoration optimizes environmental variables:

- **Water management:** Northern slopes were protected by 22 m external drains; roof repairs and precise guttering were implemented.
- **Subfloor ventilation:** A 50 cm excavation under the floor was filled with coarse river gravel and timber supports, separated from the wall by a thick tar membrane. This eliminated rising damp and maintained geothermal heat (+10–12 °C).

Outcome: During −22 °C external temperatures, interior water remained liquid. Mold disappeared without chemicals.

III. STRUCTURAL RECOVERY: BIO-REINFORCEMENT

Rather than rigid concrete, **flexible bio-frame methods** rescued leaning, cracked walls:

- **Stepwise foundation repair:** Sections of 1 m filled with compacted gravel, basalt cubes, and brick, insulated with thick tar membrane at floor level.
- **Bio-reinforcement:** 45° crossed acacia stakes and woven branch lattices at cracks transfer tensile stress, “stitching” the wall biologically rather than chemically.

IV. THE BIOPOLYMER MATRIX: FERMENTED BINDER

Adobe strength arises from microbiological processes and protein bonding. Composition:

- Recycled adobe + sand + clay
- Horse manure (~1/5 proportion)
- Whey
- Abundant straw

Fermentation mechanism: Anaerobic bacteria reduce iron oxides and partially dissolve straw cellulose to produce a natural adhesive. Within 1–2 days, cellulose bonds appear; after drying, walls gain remarkable hardness—the material sparks under a chisel.

Application: Old walls are brushed, moistened, and “primed” with thin adobe slurry before applying a 15–20 cm thick layer onto the branch lattice.

V. SURFACE TREATMENT AND VAPOR MANAGEMENT

Adobe functions as a **heat-storage and vapor-regulating system**, not a simple insulator. Layering must remain breathable:

- **Exterior layer:** Pure lime wash—vapor-permeable and self-regenerating.
- **Interior plaster:** 20 L lime slurry (sieved) + 3–4 dl plant oil + 2 kg gypsum in water. Fast-drying, smooth, and healthy indoor air quality.

VI. INNOVATION: THE ADOBE PANEL

Experiments confirmed that **whey–manure stabilized adobe panels** (4 cm thick) achieve:

- **Thermal mass:** Comparable to drywall, but naturally regulated.
- **Workability:** Can be cut and shaped like conventional construction panels.
- **Zero-waste:** Fully compostable at end-of-life.
- **Healthy:** Free from synthetic binders.

This represents the **future of eco-conscious partitioning**.

VII. HISTORICAL ADDITIONAL INSIGHTS

Ancient and medieval practices reinforce these findings:

- **Fats (water repellency):** Romans used lard (maltha), Greeks plant oils. 1–2.5 % melted fat in lime makes plaster water-repellent and glossy.
- **Proteins (strengthening):** Milk, eggs, and whey form **casein-lime**, accelerating setting and increasing decay resistance.
- **Carbohydrates:** Rice flour (10%) and honey in China, beer in the Netherlands, enhanced mechanical durability.

Modern parallel: Silane/siloxane hydrophobic treatment reduces water absorption by 90 %, with only 5–8 % reduction in vapor permeability—natural and contemporary strategies are compatible.

VIII. METHODOLOGY

1. **Field Monitoring:** Thermometers and hygrometers recorded wall and subfloor conditions during extreme winter (-22 °C).
2. **Sample Preparation:** Various mixtures of recycled adobe, sand, clay, whey, straw, and horse manure were fermented and tested for compressive and tensile strength.

3. **EFU Integration:** Environmental Flux Unit (EFU) framework mapped energy, vapor, and material flows. This informed subfloor gravel selection, wall thickness, and ventilated lime coating strategy.
4. **Iterative Testing:** Surfaces were periodically repaired, fermented panels cured under controlled temperature and humidity, and adhesion tested.
5. **Comparative Analysis:** Historical additives (fats, proteins, carbohydrates) were evaluated against modern hydrophobic treatments for performance validation.

IX. CONCLUSIONS

The house, originally completed in **1862** with walls **70–80 cm thick**, demonstrates:

- Adobe is **extremely durable and maintainable**, able to adapt to modern challenges.
- **Thermal regulation** goes beyond mass—flows of moisture and heat in the adobe matrix keep interiors temperate even during extended cold.
- **Bio-reinforcement and fermented binders** allow walls to be repaired and shaped without synthetic materials.

Legacy observation: The last inhabitants lived up to **97–98 years**, underlining the healthy living conditions provided by well-maintained adobe.

"Building with adobe does not replicate the past—it preserves the future."

X. DISCLAIMER

This study documents experimental observations and **is not a complete construction guide**. Construction must be performed by professionals.

Related Publications:

- EFU System v5.1 Diamond. DOI: 10.5281/zenodo.18888082
- MROI Preprint. DOI: 10.5281/zenodo.18766850
- *The Architecture of Hope.* DOI: 10.5281/zenodo.17793187